

Landslides

Parameters for Risk Assessments



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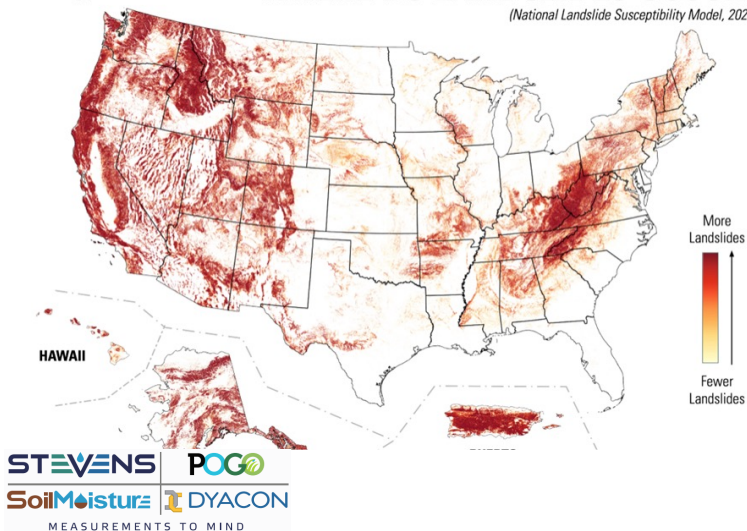


Landslides



WHERE DO LANDSLIDES OCCUR?

(National Landslide Susceptibility Model, 2024)



- Every year an estimated 400,000 landslides occur worldwide causing thousands of casualties. Landslides can pose a major threat to human lives, property and infrastructure.
- Climate change, urban growth, deforestation, and land use are considerations affecting landslide risk.
- A safety risk factor from measured parameters was developed by the United State Geological Survey (USGS) to provide an early warning alert.

Parameters Affecting Landslides

HILL SLOPE (β)

The steeper the slope, the more risk



PRECIPITATION

Rainfall saturates the soil which in turn increases the weight of the soil. Water also influences the cohesive forces in the soil.

Antecedent or Past Rainfall is used in many models.

Total Precipitation or Maximum Hourly Rainfall over the past 24 hours are often used.



VEGETATION

Vegetation promotes evapotranspiration (ET), helping the soil to dewater quickly after a precipitation event.

Intercept rainfall where water evaporates off the leaves.

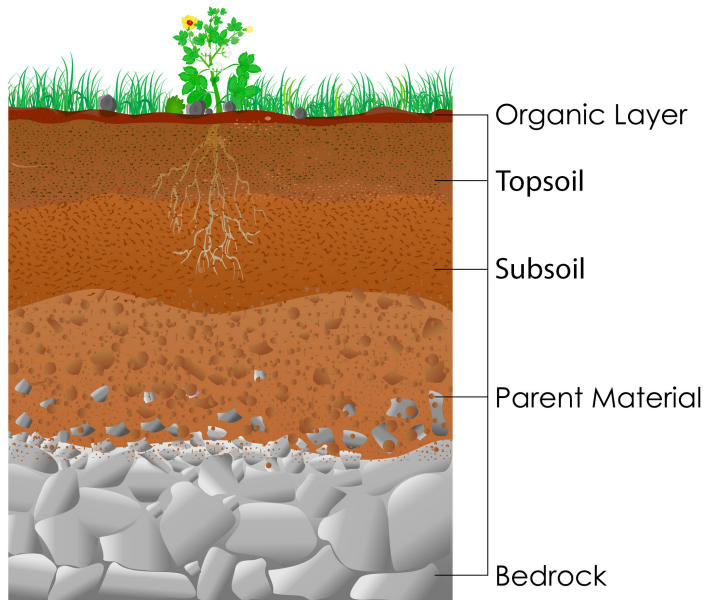
Roots can hold soil particles in place.

Vegetation index as percentage of ground cover bare soil/vegetation can be used in models



Parameters Affecting Landslides

Depth to Bedrock (H_{ss})



Biology

Soil Layer

Soil Geotechnical Parameters

Soil Stress Cohesion (c in kPa)

Soil Angle of Internal Friction (ϕ in degrees)

- Soil Stress Cohesion in units of kPa, " c "
- Soil's angle of internal friction in degrees, " ϕ "
- It represents the angle at which the material begins to fail due to shear stress. This angle is crucial in geotechnical engineering for designing foundations, slopes, and retaining structures. The angle of internal friction is determined through laboratory tests such as the direct shear test or triaxial test and is influenced by factors such as soil type, density, and moisture content.



Soil Moisture, The most important parameter

Weight of Wet Soil is influenced by soil moisture

A typical dry bulk density of soil is 1.5 grams/cubic centimeter.

Therefore one cubic meter of dry soil weighs 1500 kg.

If the soil is saturated at 45%, the weights of the water in one cubic meter of soil is 450 kg.

Under these conditions, the soil would weigh 450 kgs more per cubic meter than it would dry.

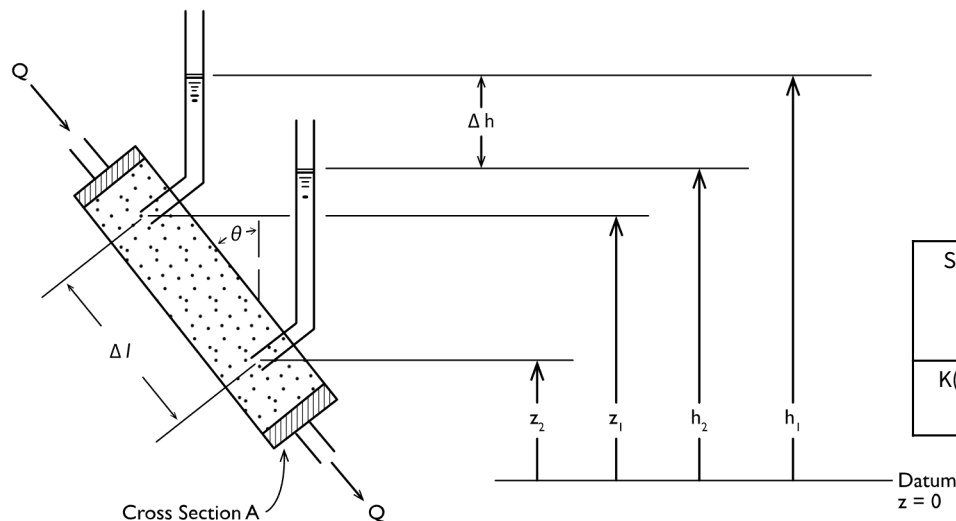
$$\rho = m/V$$

Density is equal to mass divided by volume

Soil Properties Parameters

Saturated Hydraulic Conductivity

The Saturated Hydraulic Conductivity, K_{sat} , (or K_s) represents how fast the water will drain out of soil during saturated conditions. The longer the soil stays saturated the greater the risk of slope failure.



Darcy's Law

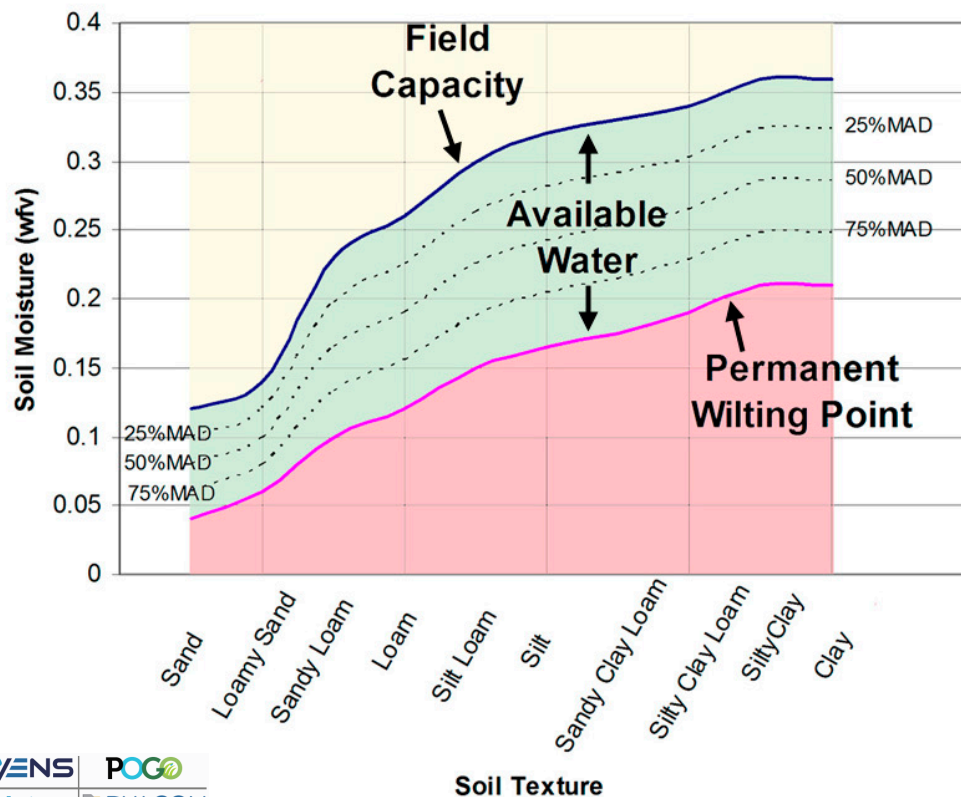
The speed of water moving through saturated soil depends on the saturated hydraulic conductivity along with the gradient of the hydraulic head with respect to distance. The K_{sat} is a property of the soil that can be measured. The bigger the K_{sat} , the faster the water will drain out.

$$v = -K_s \frac{\Delta h}{\Delta l} = -K_s \frac{dh}{dl}$$

Soil Texture	Homogeneous Clay	Structured Clay	Loam	Fine Sand	Sand	Gravel
$K(\text{SAT}) \text{ cm s}^{-1}$	10^{-8}	10^{-6}	10^{-4}	10^{-2}	1	10

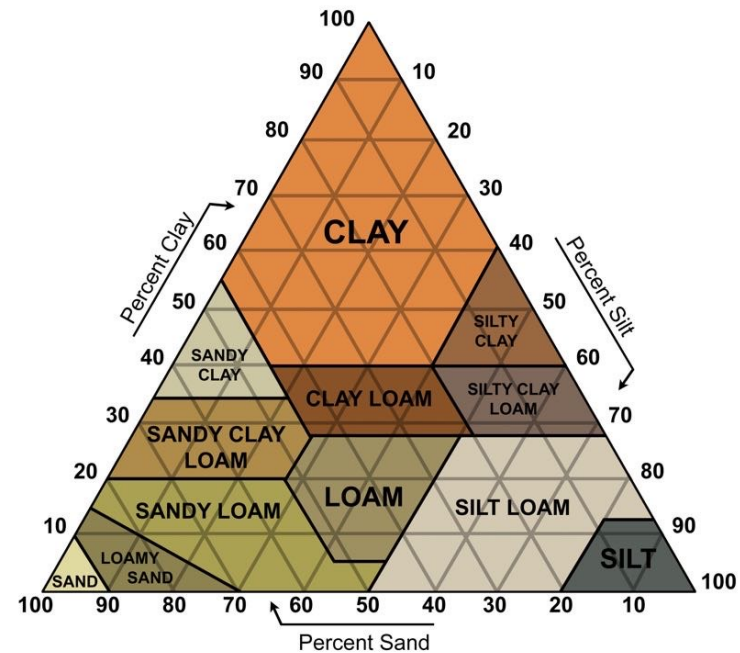
Soil Water Dynamics

Soil Moisture Target

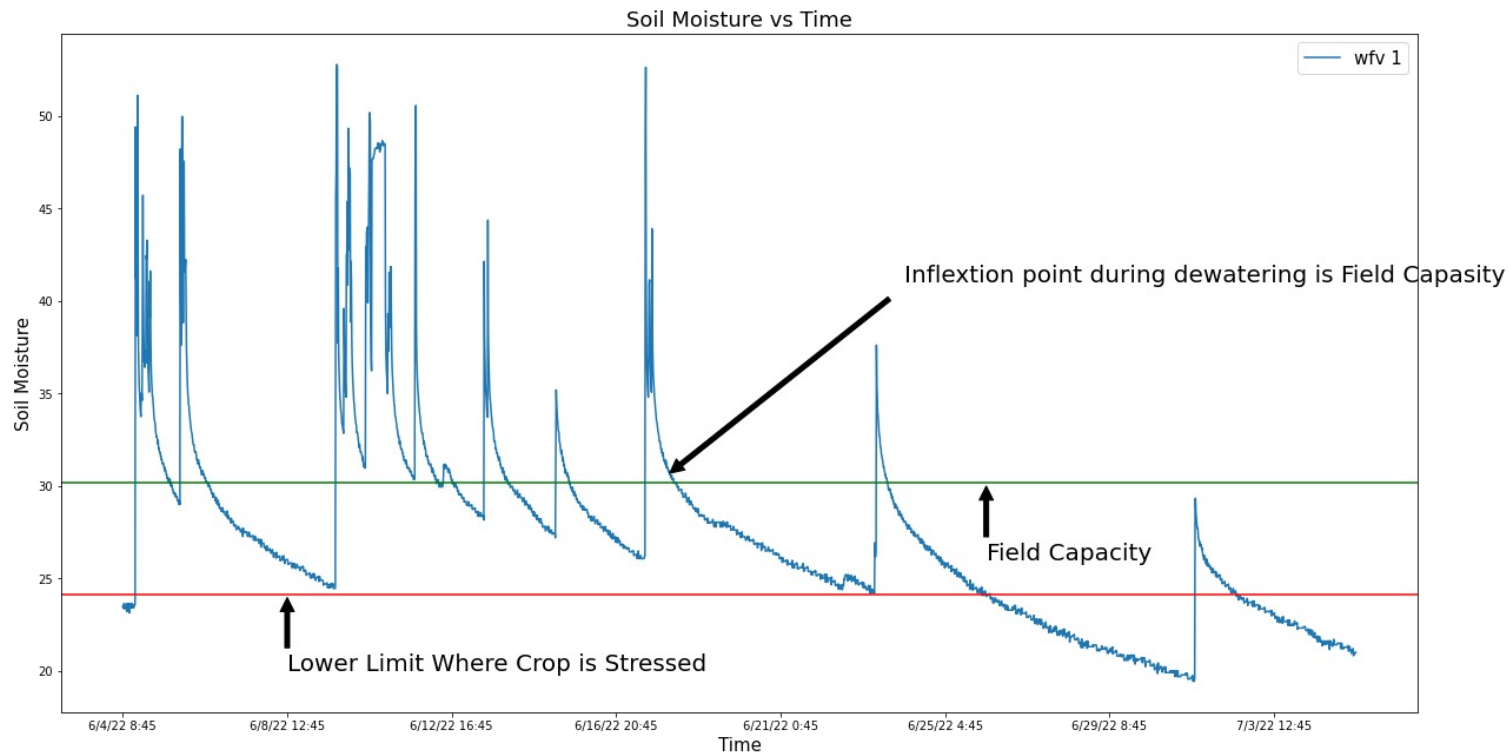


Field capacity is the point at which saturated soil drains by gravity and the water stops dripping = 0.33 Bar.

Permanent Wilting point is the moisture level where a plant can no longer pull water out, or 15 Bar.



Typical Field Capacity in mineral soil 5% Organic Matter. Soil Moisture data over a 1 month time period.



Field capacity is the inflection point in the dewatering curve. At 0.33 Bar, the movement of water switches from gravitation water flow to capillary water flow (Wicking). The movement of water from capillary forces is much slower than water draining out by gravity.

Field capacity in this time series data was 30.15 % with a standard deviation of 1.33% for 1 month. The longest it stayed at or above field capacity was 3.2 days

Groundwater

The **Saturated Zone** is the groundwater and the aquifer under the water table. It is the part of the subsurface that stays saturated all of the time. The level of the water table can be monitored in a well with a pressure sensor.

The **Unsaturated zone** (sometimes called the Vados Zone) is the part of the subsurface that is not always saturated. It is above the water table, and it is where plant roots take water out of soil. It is typically the top meter of soil.

Groundwater

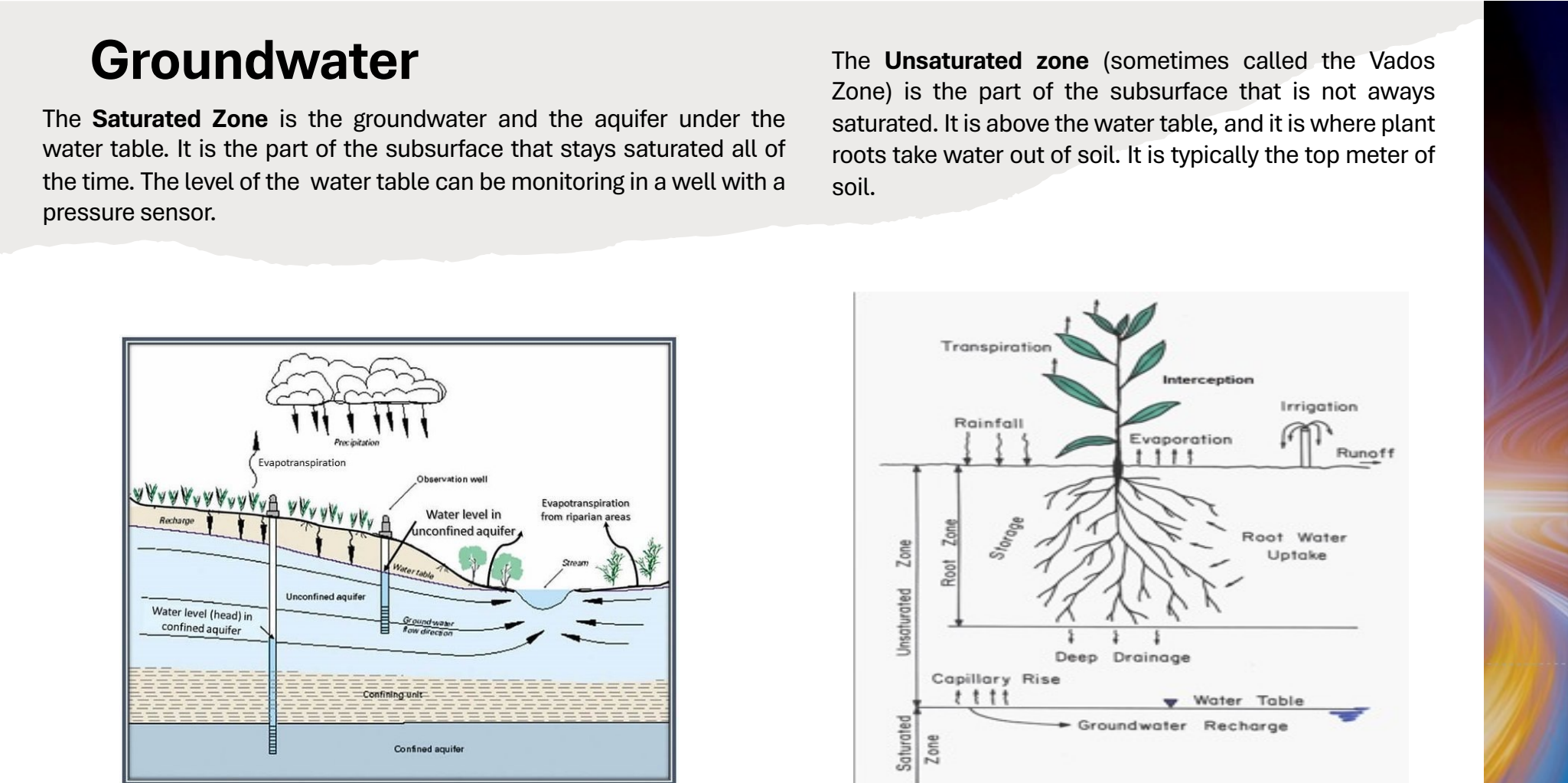
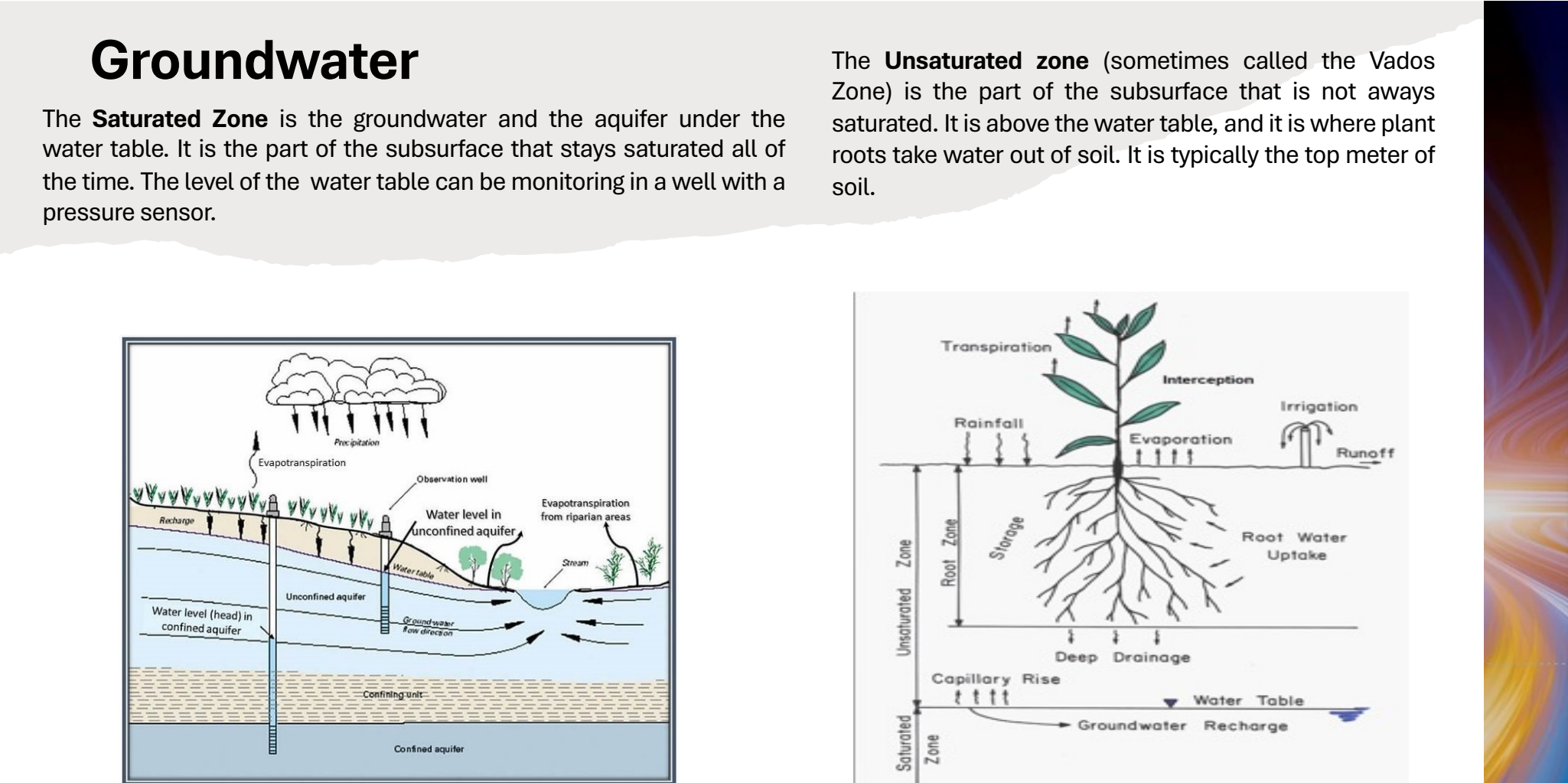
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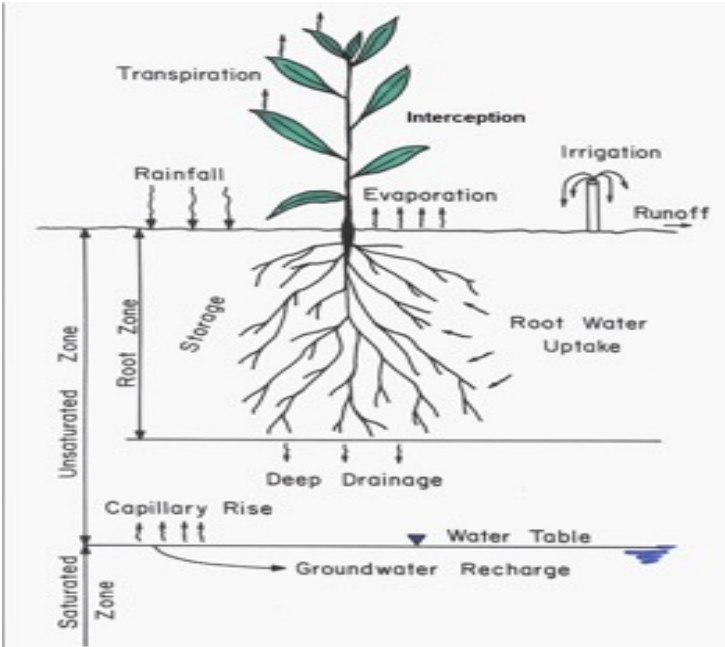
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The diagram illustrates the complex interactions between the atmosphere, soil, and subsurface. Key processes shown include:

- Atmosphere:** Precipitation, evaporation, transpiration from a plant, and interception by the canopy.
- Surface:** Rainfall, runoff, and irrigation.
- Soil:** Storage of water, root water uptake by the plant, and deep drainage.
- Subsurface:** The unsaturated zone (root zone) and the saturated zone (water table). Processes include recharge, groundwater flow direction, and capillary rise.

Figure 5.1 Transient water flow processes in soil (Šimůnek and van Genuchten, 2007).

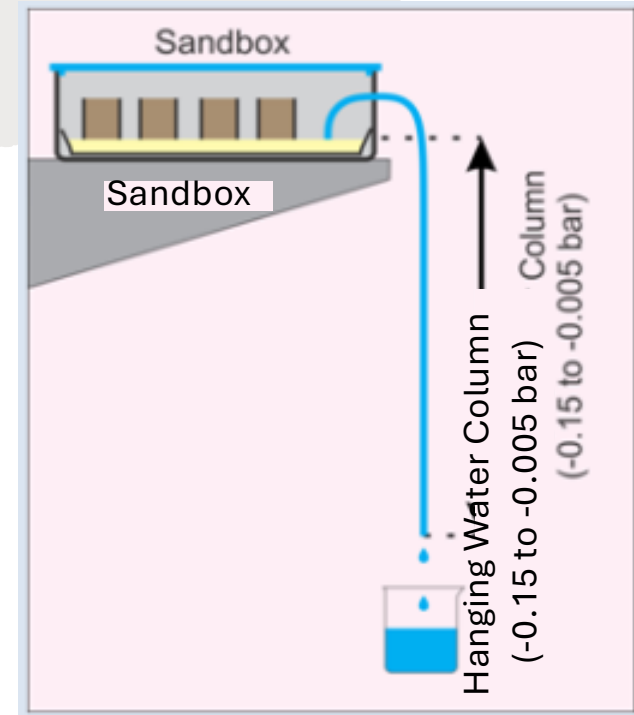
Soil Water Potential ψ

Matric Potential is the amount of pressure it takes to pull water out of soil.

ψ_m is the matric potential in units of negative pressure such as Pa.

$$\psi_m = -\rho_w g H$$

Words that are used to describe Matric Potential	Units
Matric Potential	Pressure
Soil Suction	Pressure
Soil Tension	Pressure
Hydraulic Head	Normally expressed as meters
pF	Log of matric potential in hPa

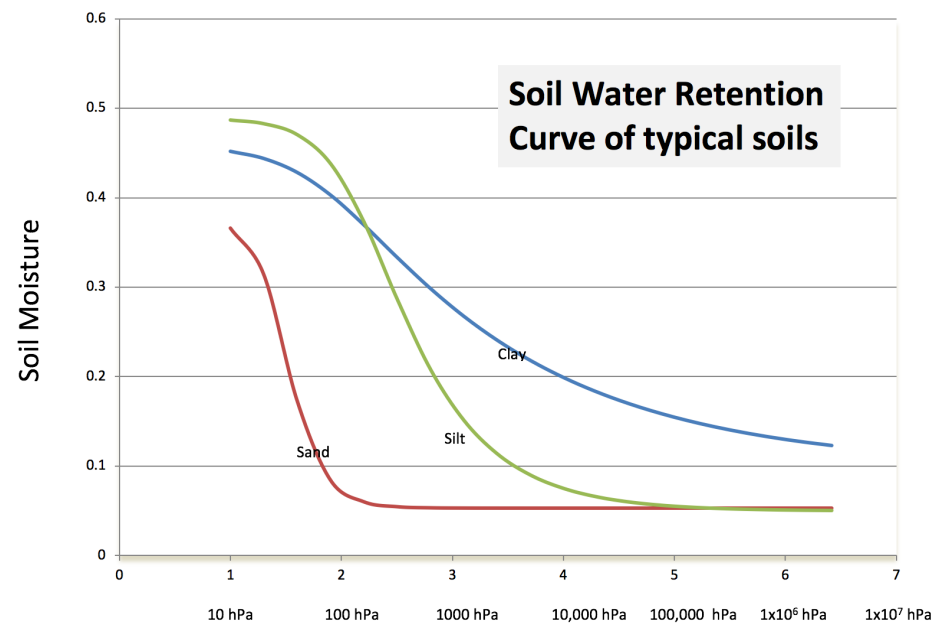


Soil Properties Parameters

Soil Water Retention Curve

The soil water retention curve is a plot of soil moisture on the y axis and matric potential (pressure it take to pull water out of soil) on the x axis.

The soil water retention curve is very soil specific and can help determine how water drains and moves through he soil.



van Genuchten Equation

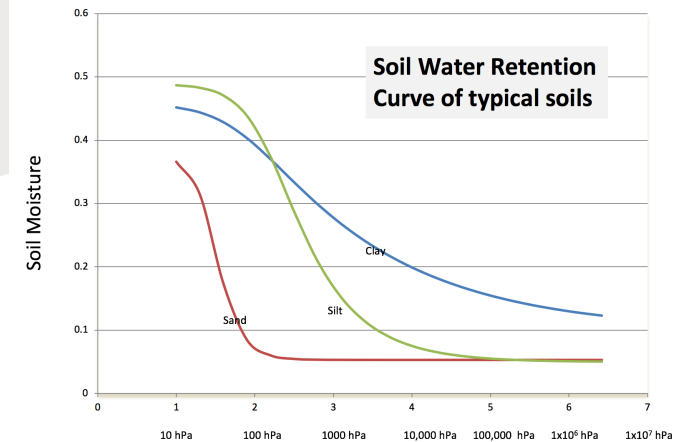
Water Content Modeling and Prediction

$$\theta(h) = \theta_r + \frac{\theta_s - \theta_r}{[1 + \alpha |h|^n]^m}$$

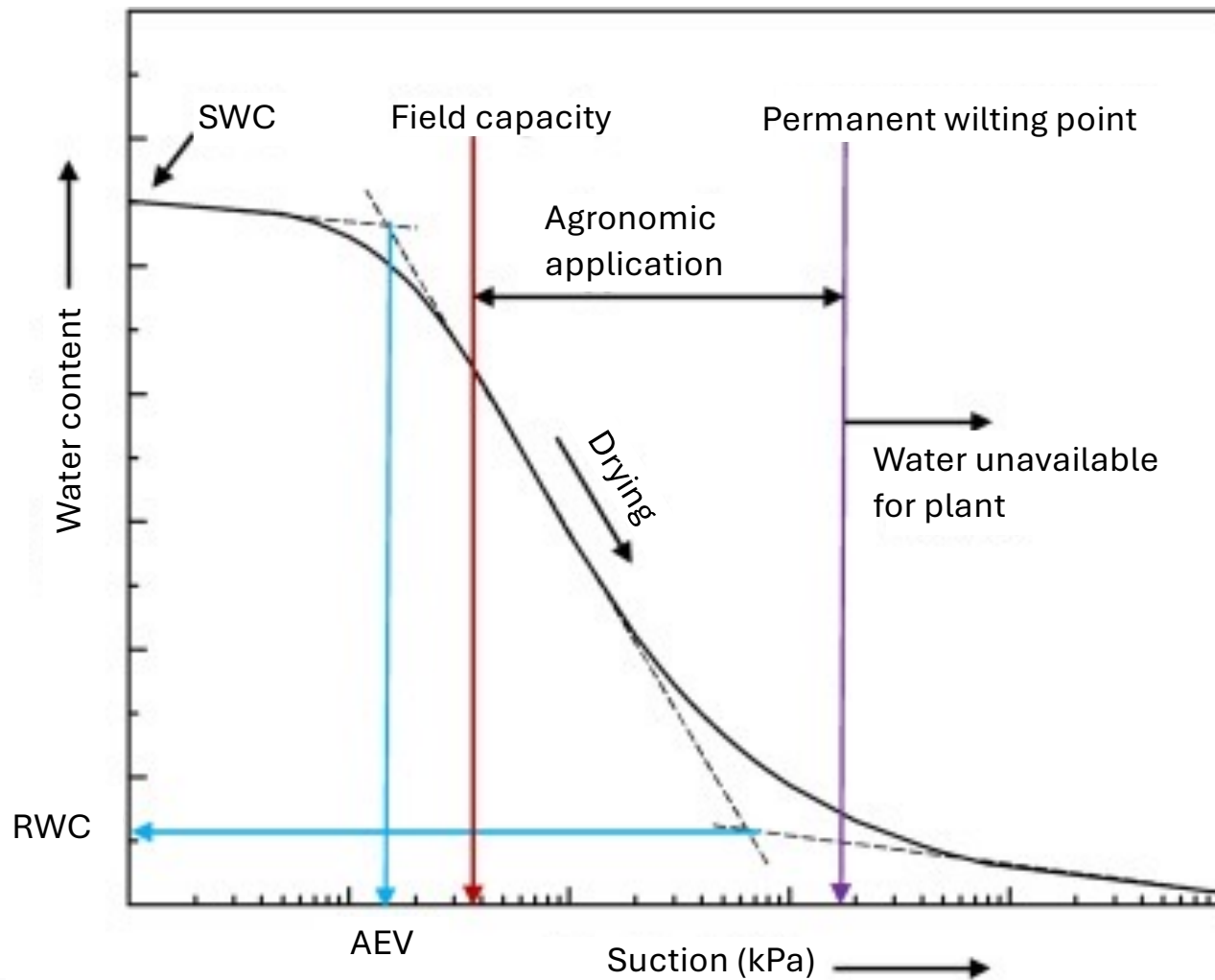
$$S_e(h) = \frac{\theta(h) - \theta_r}{\theta_s - \theta_r}$$

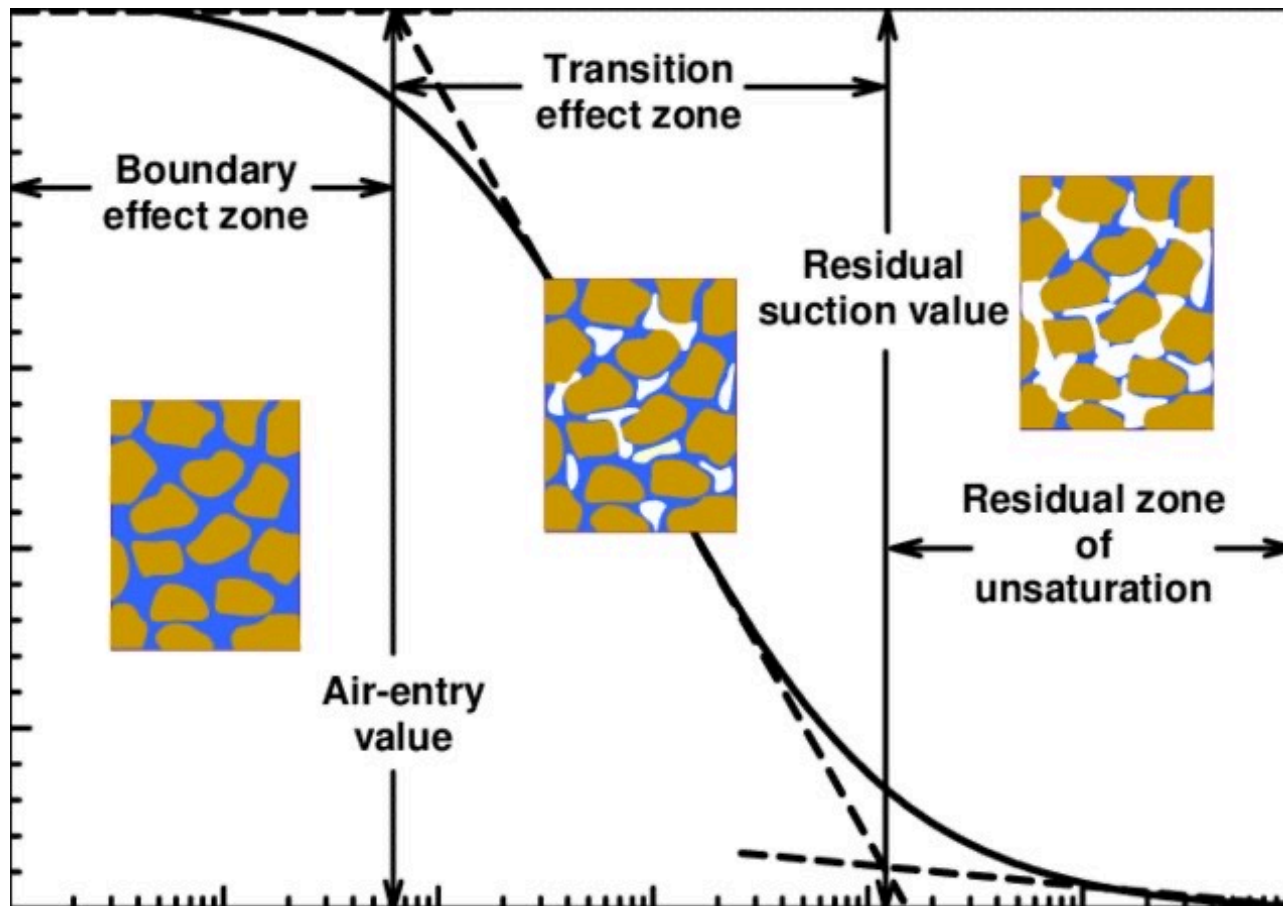
$$K(h) = K_s S_e^{\ell} \left[1 - \left(1 - S_e^{1/m} \right)^m \right]^2$$

$K(h)$ = The Unsaturated Hydraulic Conductivity
[a parameter that indicates how well water moves through soil]



Symbol	Definition - Định nghĩa
θ_r	Soil water residual. A low level of soil moisture where the microscopic channels of water in soil become disconnected.
θ_s	Soil moisture at saturation.
α	Air entry value. Point below saturation where the air in soil becomes displaced.
n	Fitting parameter that is soil specific.
h	Hydraulic head which is basically the same thing as the matric potential.





Where to Place a Monitoring Station

Parameter Station Placement

Hill Slope Angle

Depth to Bed Rock

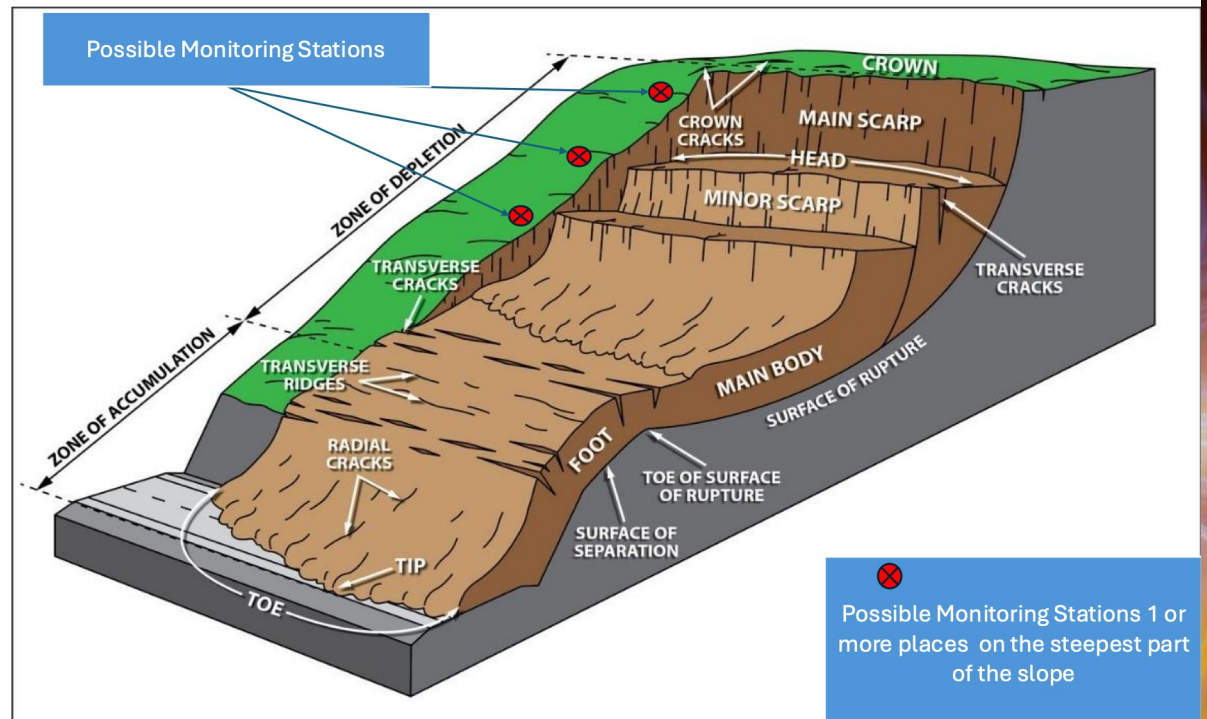
Vegetation

Human Habitation and
property

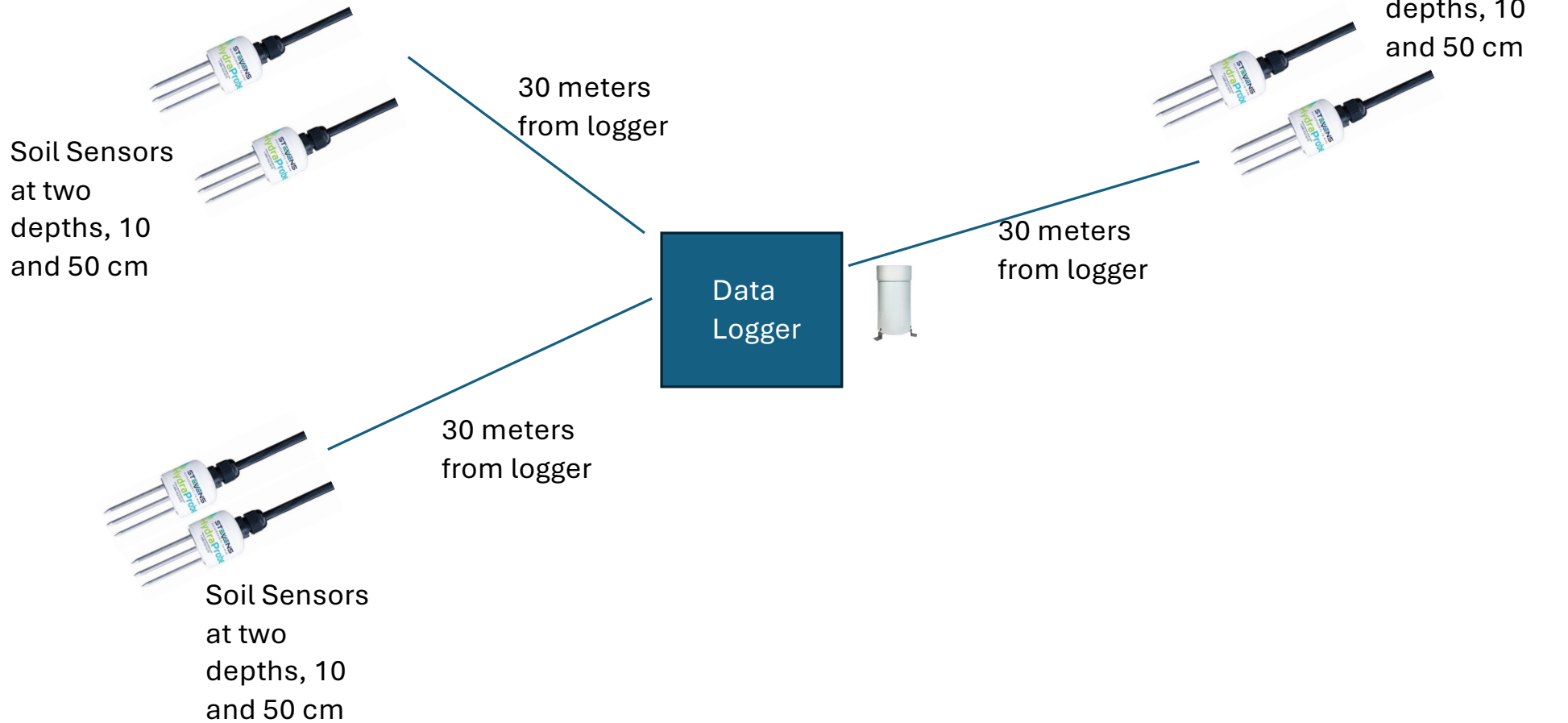
Saturated Hydrologic
conductivity

Soil Stress Cohesion

Soil's angle of internal
friction



Possible Monitoring Station



Landslide Risk Factor

Developed by The US Geological Survey

$$FS = \frac{\tan(\phi)}{\tan(\beta)} + \frac{2c}{\gamma H_{ss} \sin(2\beta)} + \frac{\psi}{\gamma H_{ss}} [\tan(\beta) + \cot(\beta)] \tan(\phi)$$



Variable	Definition	Remarks
FS	Landslide safety factor	Reaches a minimum when Landslide occurs
ϕ	Friction Angle	Measured in lab or estimated
β	Hill Slope Angle	
c	Soil Cohesion	Measured in lab or estimated
γ	Soil unit Weight	Measured in lab or estimated, or 17.28 kN/m ³ for most soils
H _{ss}	Depth to bedrock	Measured with drilling or estimated from geological survey
ψ	Hydraulic head (Matric Potential)	Measure with a soil moisture sensor and a water retention curve

Application of Landslide Risk Factor

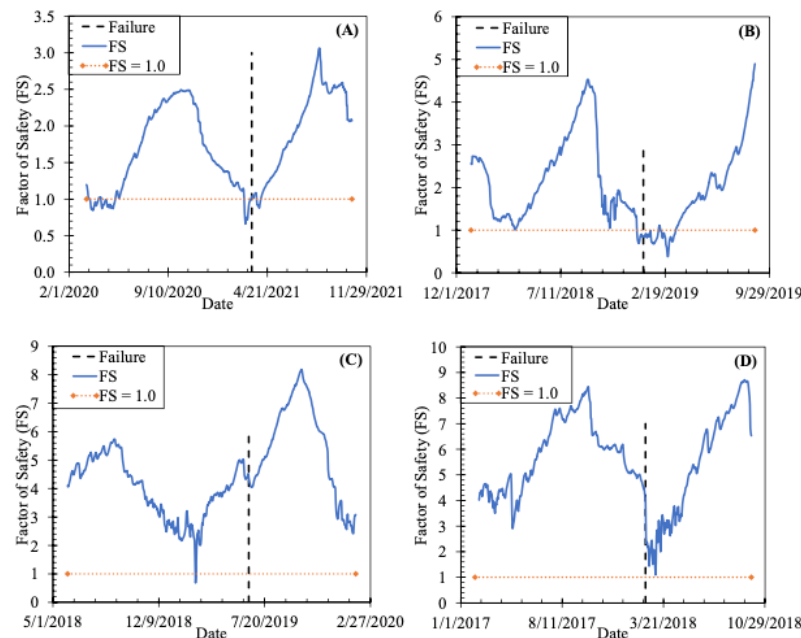


Fig. 8 Infinite slope stability analysis at: **A** Landslide 10,757, Occurred 03/17/2021, **B** Landslide 8596, Reported 01/04/2019, **C** Landslide 9677, Occurred 06/17/2019, and **D** Landslide 6456, Occurred 02/10/2018

$$FS = \frac{\tan(\phi)}{\tan(\beta)} + \frac{2c}{\gamma H_{ss} \sin(2\beta)} + \frac{\psi}{\gamma H_{ss}} [\tan(\beta) + \cot(\beta)] \tan(\phi)$$

Parameter Monitoring and determination of Safety Factor

Hill Slope Angle

Depth to Bed Rock

Soil Moisture

Soil/water retention curve

Soil Stress Cohesion

Soil's angle of internal friction

Advanced Landslide Risk Factor Over a Region

Natural Hazards

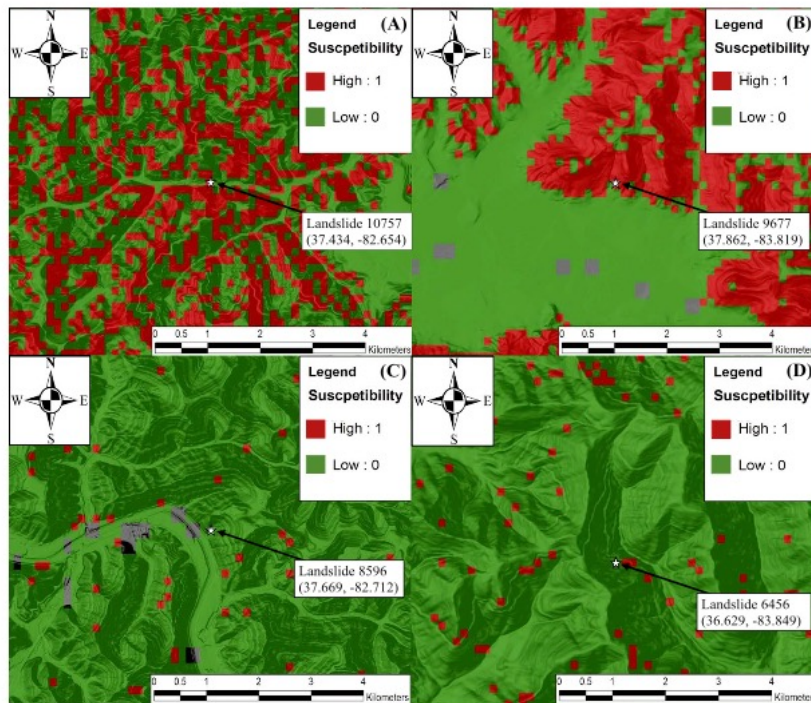


Fig.7 Susceptibility over: **A** Landslide 10,757, Occurred 03/17/2021, **B** Landslide 9677, Occurred 06/17/2019, **C** Landslide 8596, Reported 01/04/2019, and **D** Landslide 6456, Occurred 02/10/2018

(Francis, 2024) Study Conducted by the University of Kentucky in the United States.

Machine Learning and Satellite Remote Sensing Data can be used to find the safety factor over an area in near real time.

Onsite Equipment

WATER RETENTION CURVES

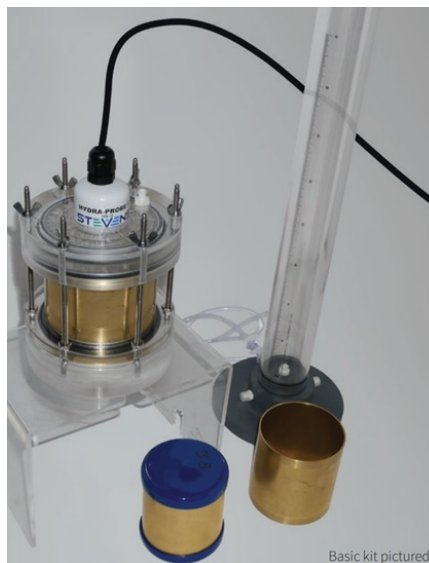
Unsaturated Hydraulic Conductivity $K(h)$

Soil moisture at 0.33 bar.



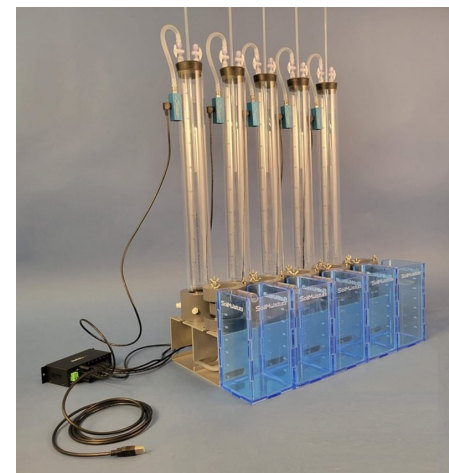
TEMPE CELL

Porosity and dry bulk density.



FALLING HEAD PERMEAMETER

Saturated Hydraulic Conductivity (K_s) and Unsaturated Hydraulic Conductivity (K_h).



Onsite Equipment

HydraProbe SOIL SENSOR

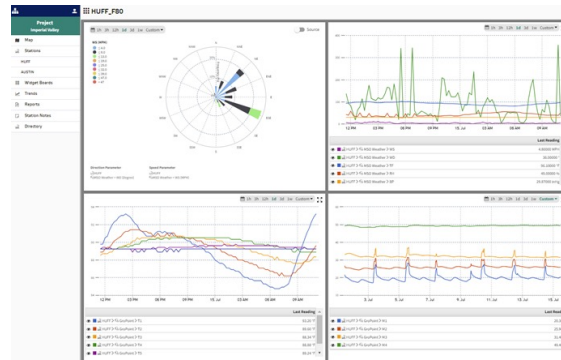
Measures soil moisture.

Connects to datalogger.



STEELHEAD DATALOGGER + SKYVIEW 360

Monitoring dashboard → Early warning.



Optional Sensors

- Rain Fall
- Groundwater level if there are monitoring wells.



Conclusion

Landslide Monitoring Station

- Geologic Parameters such as Slope Angle can be used to determine Placement of a Monitoring Station.
- Monitoring Station;
 - Geologic Parameters can be measured in a lab or estimated.
 - Soil Moisture can be measured with 6 to 10 soil moisture sensors
 - A safety factor can be determined from soil moisture data and geologic parameters

Safety Factor and
Sensor Data can be
uploaded to the
cloud

Data and
calculation can
be viewed and
managed in Sky
View 360



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Thank You!



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